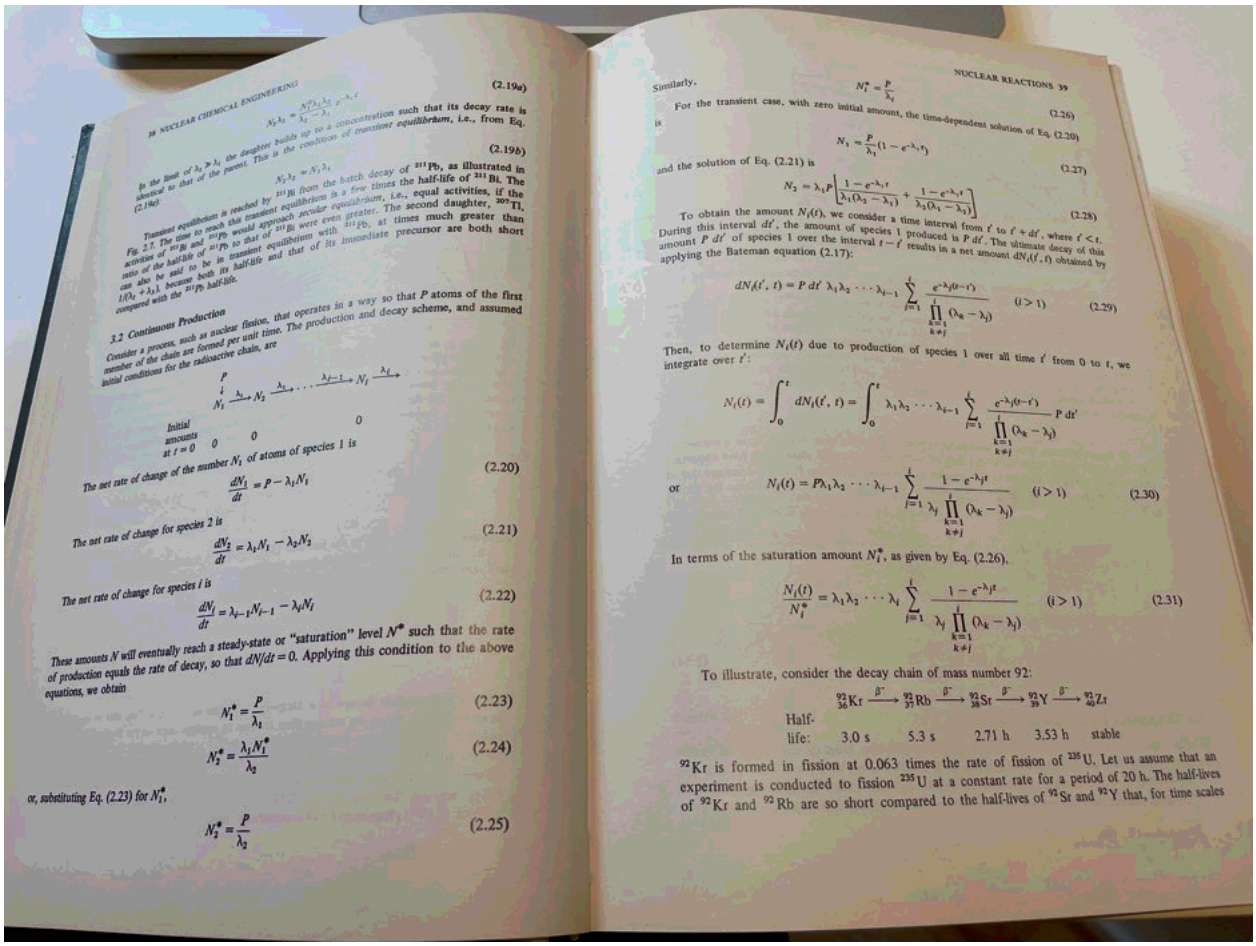
1. Half-life (finished)
2. Decay chain (finished)
3. Continuous production

Similar to the decay chain example, but now have a slider bar for P, the course of species 1, and only have three species (two decay constants) instead of 4.



4. Continuous production with shutdown

Same as previous, but now hold P fixed and have slider bars for the two decay constants and the time at which P is switched to zero

5. Continuous production with continuous removal.

Here we have a decay chain of three species, and a fixed P, and we will remove species 2 from the system with rate f. f should have a slider bar, along with the two decay constants.

Table 2.13 Ratio of fission-product decay-heat rate with neutron absorption to decay-heat rate without absorption†

Time after reactor shutdown, s	G_{max}	Time after reactor shutdown, s	G_{max}
—	1.020	6.0×10^4	1.111
1.0	1.020	8.0×10^4	1.119
1.5	1.021	1.0×10^5	1.124
2.0	1.022	1.5×10^5	1.130
4.0	1.022	2.0×10^5	1.131
8.0	1.022	4.0×10^5	1.126
1.0×10^1	1.022	6.0×10^5	1.124
1.5×10^1	1.022	8.0×10^5	1.123
2.0×10^1	1.022	1.0×10^6	1.124
4.0×10^1	1.022	1.5×10^6	1.125
6.0×10^1	1.022	2.0×10^6	1.127
8.0×10^1	1.023	4.0×10^6	1.134
1.0×10^2	1.024	6.0×10^6	1.146
1.5×10^2	1.025	8.0×10^6	1.162
2.0×10^2	1.028	1.0×10^7	1.181
4.0×10^2	1.030	1.5×10^7	1.233
6.0×10^2	1.032	2.0×10^7	1.284
8.0×10^2	1.033	4.0×10^7	1.444
1.0×10^3	1.037	6.0×10^7	1.535
1.5×10^3	1.039	8.0×10^7	1.586
2.0×10^3	1.048	1.0×10^8	1.598
4.0×10^3	1.054	1.5×10^8	1.498
6.0×10^3	1.060	2.0×10^8	1.343
8.0×10^3	1.064	4.0×10^8	1.065
1.0×10^4	1.074	6.0×10^8	1.021
1.5×10^4	1.081	8.0×10^8	1.012
2.0×10^4	1.098	1.0×10^9	1.007
4.0×10^9			

†Ratio based on ^{235}U thermal fission for 4 years, no depletion, typical spectrum for light-water reactor.

Source: American Nuclear Society Standards Committee Working Group ANS-5.1, "American National Standard for Decay Heat Power in Light Water Reactors," Standard ANSI/ANS-5.1, American Nuclear Society, La Grange Park, Ill., 1979. With permission of the publisher, the American Nuclear Society.

and by continuous processing. Examples of continuous-processing removal are the vaporization of one or more gaseous elements from a solid or liquid at high temperature or the continuous separation of one or more chemical elements from a well-stirred fluid mixture. Nuclides within the chain under consideration are linked by radioactive decay or neutron reactions. In the present analysis for batch operation we assume that there is a finite initial amount of only the first member of the chain, that there is no source for continuous formation of this first member, and that there is no source of any other member of the chain other than its precursor in the chain itself.

First we assume a chain in which adjacent members are linked by radioactive decay. The

neutron flux and reaction cross sections are assumed to be time-independent. The chain and the removal-rate constants are indicated schematically below:

Chain:	$N_1 \xrightarrow[\downarrow \phi\sigma_1 + f_1]{\lambda_1}$	$N_2 \xrightarrow[\downarrow \phi\sigma_2 + f_2]{\lambda_2}$	$N_3 \xrightarrow[\downarrow \phi\sigma_3 + f_3]{\lambda_3}$	$N_j \xrightarrow[\downarrow \phi\sigma_j + f_j]{\lambda_j} \cdots \xrightarrow{\lambda_{i-1}}$	$N_i \xrightarrow[\downarrow \phi\sigma_i + f_i]{\lambda_i}$
Initial amount at $t = 0$	N_1^0	0	0	0	0
Removal-rate constants:					
Radioactive decay	λ_1	λ_2	λ_3	λ_j	λ_i
Neutron absorption	$\phi\sigma_1$	$\phi\sigma_2$	$\phi\sigma_3$	$\phi\sigma_j$	$\phi\sigma_i$
Continuous processing	f_1	f_2	f_3	f_j	f_i
Total	μ_1	μ_2	μ_3	μ_j	μ_i

The processing removal constant f is interpreted as the fraction removed per unit time by continuous reprocessing. The total removal-rate constant μ is defined as

$$\mu = \lambda + \phi\sigma + f \tag{2.95}$$

In the absence of sources, the time-dependent equation for the number of atoms N_1 of the first member of the chain is

$$\frac{dN_1}{dt} = -(\lambda_1 + \phi\sigma_1 + f_1)N_1 \tag{2.96}$$

or, from Eq. (2.95),

$$\frac{dN_1}{dt} = -\mu_1 N_1 \tag{2.97}$$

For time-independent μ, Eq. (2.97) integrates to

$$N_1 = N_1^0 e^{-\mu_1 t} \tag{2.98}$$

Similarly, for the second member,

$$\frac{dN_2}{dt} = \lambda_1 N_1 - \mu_2 N_2 \tag{2.99}$$

which integrates to

$$N_2 = N_1^0 \lambda_1 \left(\frac{e^{-\mu_1 t}}{\mu_2 - \mu_1} + \frac{e^{-\mu_2 t}}{\mu_1 - \mu_2} \right) \tag{2.100}$$

Similarly, using the technique described in Sec. 7, the general batch equation is obtained:

$$N_1 = N_1^0 e^{-\mu_1 t} \tag{2.101a}$$

$$N_i = N_1^0 \lambda_1 \lambda_2 \cdots \lambda_{i-1} \sum_{j=1}^{i} \frac{e^{-\mu_j t}}{\prod_{\substack{k=1 \\ k \neq j}}^{i} (\mu_k - \mu_j)} \qquad (i > 1) \tag{2.101b}$$

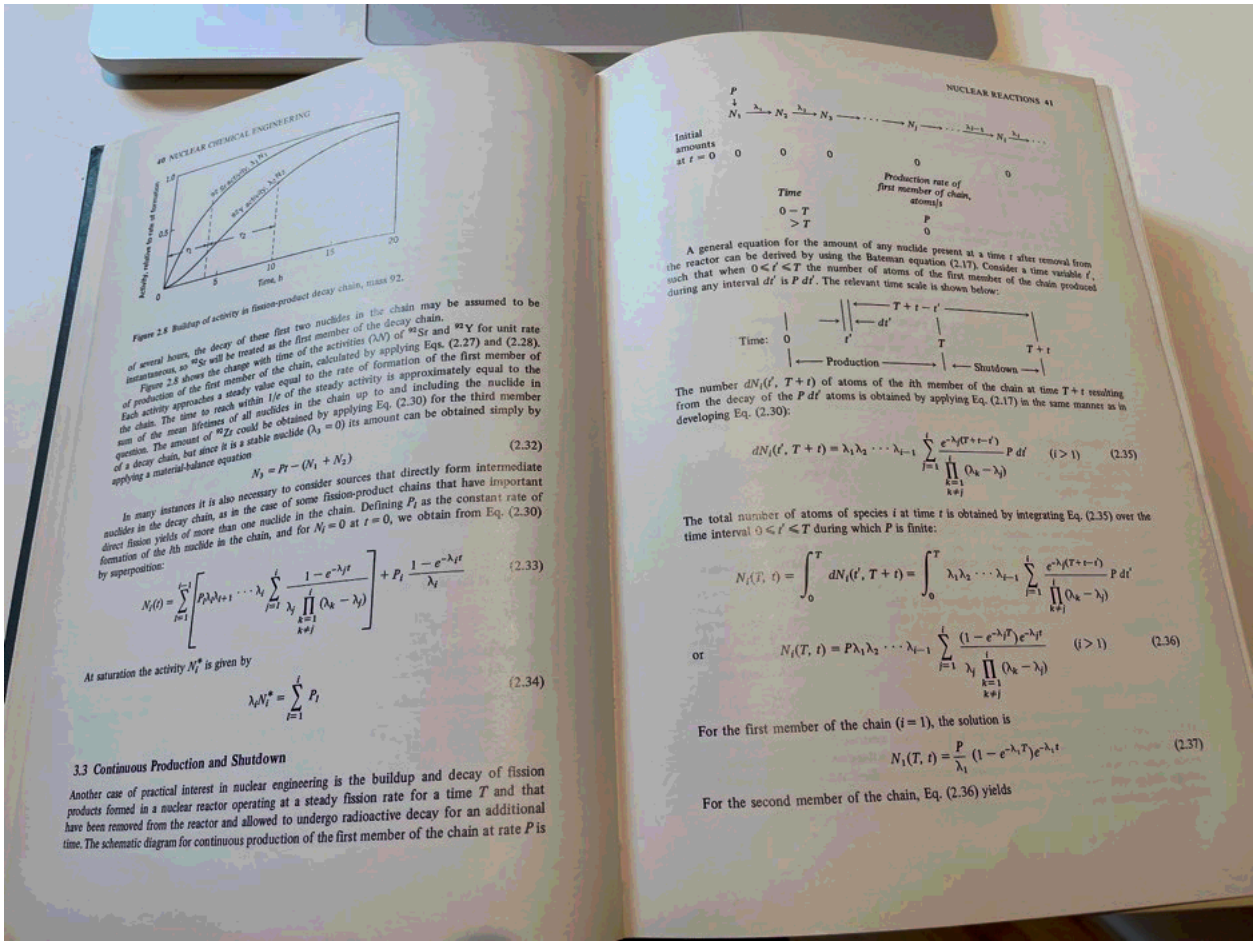


Figure 2.8 Buildup of activity in fission-product decay chain, mass 92.

of several hours, the decay of these first two nuclides in the chain may be assumed to be instantaneous, so ^{92}Sr will be treated as the first member of the decay chain.

Figure 2.8 shows the change with time of the activities (λN) of ^{92}Sr and ^{92}Y for unit rate of production of the first member of the chain, calculated by applying Eqs. (2.27) and (2.28). Each activity approaches a steady value equal to the rate of formation of the first member of the chain. The time to reach within 1/e of the steady activity is approximately equal to the sum of the mean lifetimes of all nuclides in the chain up to and including the nuclide in question. The amount of ^{92}Zr could be obtained by applying Eq. (2.30) for the third member of a decay chain, but since it is a stable nuclide ($\lambda_3 = 0$) its amount can be obtained simply by applying a material-balance equation

$$N_3 = Pt - (N_1 + N_2) \tag{2.32}$$

In many instances it is also necessary to consider sources that directly form intermediate nuclides in the decay chain, as in the case of some fission-product chains that have important direct fission yields of more than one nuclide in the chain. Defining P_i as the constant rate of formation of the ith nuclide in the chain, and for $N_i = 0$ at $t = 0$, we obtain from Eq. (2.30) by superposition:

$$N_i(t) = \sum_{j=1}^{i-1}\left[P_j\lambda_{j+1}\cdots\lambda_i\sum_{j=1}^{i}\frac{1-e^{-\lambda_j t}}{\lambda_j\prod_{\substack{k=1\\k\neq j}}^{i}(\lambda_k-\lambda_j)}\right] + P_i\frac{1-e^{-\lambda_i t}}{\lambda_i} \tag{2.33}$$

At saturation the activity N_i^* is given by

$$\lambda_i N_i^* = \sum_{j=1}^{i} P_j \tag{2.34}$$

3.3 Continuous Production and Shutdown

Another case of practical interest in nuclear engineering is the buildup and decay of fission products formed in a nuclear reactor operating at a steady fission rate for a time T and that have been removed from the reactor and allowed to undergo radioactive decay for an additional time. The schematic diagram for continuous production of the first member of the chain at rate P is

A general equation for the amount of any nuclide present at a time t after removal from the reactor can be derived by using the Bateman equation (2.17). Consider a time variable t', such that when $0 \leq t' \leq T$ the number of atoms of the first member of the chain produced during any interval dt' is $P\,dt'$. The relevant time scale is shown below:

The number $dN_i(t', T+t)$ of atoms of the ith member of the chain at time $T+t$ resulting from the decay of the $P\,dt'$ atoms is obtained by applying Eq. (2.17) in the same manner as in developing Eq. (2.30):

$$dN_i(t', T+t) = \lambda_1\lambda_2\cdots\lambda_{i-1}\sum_{j=1}^{i}\frac{e^{-\lambda_j(T+t-t')}}{\prod_{\substack{k=1\\k\neq j}}^{i}(\lambda_k-\lambda_j)}P\,dt' \qquad (i>1) \tag{2.35}$$

The total number of atoms of species i at time t is obtained by integrating Eq. (2.35) over the time interval $0 \leq t' \leq T$ during which P is finite:

$$N_i(T, t) = \int_0^T dN_i(t', T+t) = \int_0^T \lambda_1\lambda_2\cdots\lambda_{i-1}\sum_{j=1}^{i}\frac{e^{-\lambda_j(T+t-t')}}{\prod_{\substack{k=1\\k\neq j}}^{i}(\lambda_k-\lambda_j)}P\,dt'$$

or

$$N_i(T, t) = P\lambda_1\lambda_2\cdots\lambda_{i-1}\sum_{j=1}^{i}\frac{(1-e^{-\lambda_j T})e^{-\lambda_j t}}{\lambda_j\prod_{\substack{k=1\\k\neq j}}^{i}(\lambda_k-\lambda_j)} \qquad (i>1) \tag{2.36}$$

For the first member of the chain ($i = 1$), the solution is

$$N_1(T, t) = \frac{P}{\lambda_1}(1-e^{-\lambda_1 T})e^{-\lambda_1 t} \tag{2.37}$$

For the second member of the chain, Eq. (2.36) yields

6. Chain reaction

A + n -> B + C + X n

Second order reaction:

dA/dt = -k*A*n

dn/dt = -k*A*n + k*A*n*X = k*(X-1)*A*n

Slider bars for k and X (0 to ~3)

Can use the example below with U-235 fission:

chemical name or as a superscript preceding the chemical symbol; thus, the heaviest isotope of uranium would be represented as uranium-238, or ^{238}U.

The complete notation for a nuclide is ${}^{A}_{Z}$(element symbol). For example, ${}^{85}_{36}$Kr is the isotope of krypton of mass number 85 ($Z = 36$, $A = 85$). Use of the atomic number in this symbol is redundant because all isotopes of an element have the same Z, but it is convenient in balancing equations for nuclear reactions.

In the published charts of the nuclides and in the compilation in App. C, the atomic masses are listed in physical mass units (amu), in which one atom of ^{12}C has a mass of 12.0000000.

Some nuclei with a given A and Z can exist temporarily in metastable states having more energy than the ground state, corresponding to that of A and Z. Nuclei with the same A and the same Z, but different energies, are called *isomers*. The higher-energy ones are represented by placing an m or * after the mass number, for example, ${}^{85m}_{36}$Kr.

1.2 Balancing Nuclear Reactions

It is characteristic of nuclear reactions of the type occurring in nuclear reactors that the sum of the number of neutrons and protons in the reactants equals the sum in the products. The same is true of the charge of the reactants and products. Consequently, in balancing nuclear reactions, the sum of the A's of the reactants must equal the sum of the A's of the products; and the sum of the Z's of the reactants must equal the sum of the Z's of the products. As an example of a balanced equation for a nuclear reaction, we may consider one of the fission reactions that occurs when ^{235}U absorbs a neutron:

$$^{235}_{92}\mathrm{U} + {}^{1}_{0}n \rightarrow {}^{144}_{54}\mathrm{Xe} + {}^{89}_{38}\mathrm{Sr} + 3{}^{1}_{0}n$$

The neutron is represented by ${}^{1}_{0}n$, a nuclide with nuclear charge 0 and mass number 1.

2 RADIOACTIVITY

2.1 Types

Radioactive nuclides break down spontaneously in six principal ways, illustrated by the following examples:

1. Alpha decay:

$$^{239}_{94}\mathrm{Pu} \rightarrow {}^{235}_{92}\mathrm{U} + {}^{4}_{2}\mathrm{He} \quad \text{(alpha particle)}$$

2. Beta decay:

$$^{89}_{38}\mathrm{Sr} \rightarrow {}^{89}_{39}\mathrm{Y} + {}^{0}_{-1}e \quad \text{(electron)}$$

3. Gamma emission:

$$^{85m}_{36}\mathrm{Kr} \rightarrow {}^{85}_{36}\mathrm{Kr} + {}^{0}_{0}\gamma \quad \text{(gamma photon)}$$

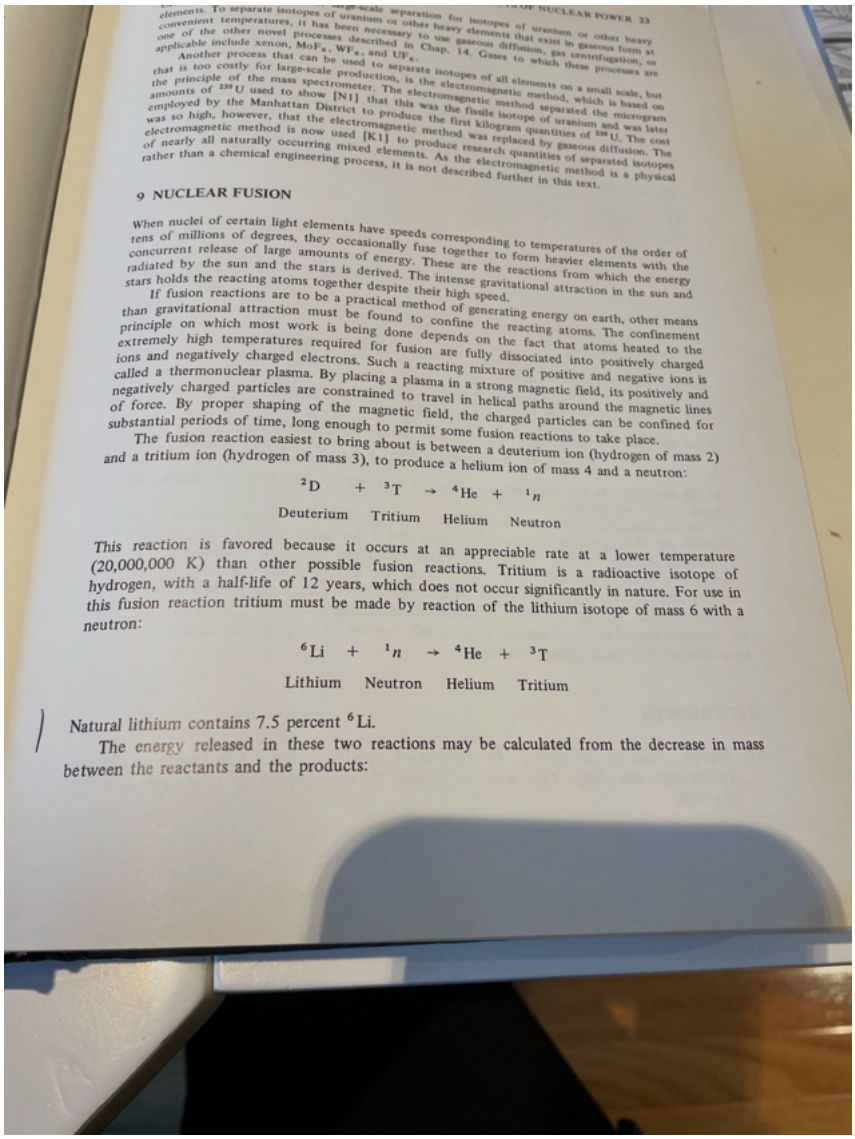
4. Positron emission:

$$^{83}_{38}\mathrm{Sr} \rightarrow {}^{83}_{37}\mathrm{Rb} + {}^{0}_{1}e \quad \text{(positron)}$$

$$^{0}_{1}e + {}^{0}_{-1}e \rightarrow 2{}^{0}_{0}\gamma \quad \text{(0.51 MeV photons)}$$

7. Fusion

Energy released is 17.6 MeV per pair of atoms fused, which is 2.69e14 J. Vary the initial amounts of deuterium and tritium and calculate the species concentrations (second order reaction) and integrated energy released over time.



elements. To separate isotopes of uranium or other heavy
convenient temperatures, it has been necessary to use gaseous diffusion, gas centrifugation, or
one of the other novel processes described in Chap. 14. Gases to which these processes are
applicable include xenon, MoF_6, WF_6, and UF_6.

Another process that can be used to separate isotopes of all elements on a small scale, but
that is too costly for large-scale production, is the electromagnetic method, which is based on
the principle of the mass spectrometer. The electromagnetic method separated the microgram
amounts of ^{235}U used to show [N1] that this was the fissile isotope of uranium and was later
employed by the Manhattan District to produce the first kilogram quantities of ^{235}U. The cost
was so high, however, that the electromagnetic method was replaced by gaseous diffusion. The
electromagnetic method is now used [K1] to produce research quantities of separated isotopes
of nearly all naturally occurring mixed elements. As the electromagnetic method is a physical
rather than a chemical engineering process, it is not described further in this text.

9 NUCLEAR FUSION

When nuclei of certain light elements have speeds corresponding to temperatures of the order of tens of millions of degrees, they occasionally fuse together to form heavier elements with the concurrent release of large amounts of energy. These are the reactions from which the energy radiated by the sun and the stars is derived. The intense gravitational attraction in the sun and stars holds the reacting atoms together despite their high speed.

If fusion reactions are to be a practical method of generating energy on earth, other means than gravitational attraction must be found to confine the reacting atoms. The confinement principle on which most work is being done depends on the fact that atoms heated to the extremely high temperatures required for fusion are fully dissociated into positively charged ions and negatively charged electrons. Such a reacting mixture of positive and negative ions is called a thermonuclear plasma. By placing a plasma in a strong magnetic field, its positively and negatively charged particles are constrained to travel in helical paths around the magnetic lines of force. By proper shaping of the magnetic field, the charged particles can be confined for substantial periods of time, long enough to permit some fusion reactions to take place.

The fusion reaction easiest to bring about is between a deuterium ion (hydrogen of mass 2) and a tritium ion (hydrogen of mass 3), to produce a helium ion of mass 4 and a neutron:

$$^2D + {}^3T \rightarrow {}^4He + {}^1n$$

Deuterium Tritium Helium Neutron

This reaction is favored because it occurs at an appreciable rate at a lower temperature (20,000,000 K) than other possible fusion reactions. Tritium is a radioactive isotope of hydrogen, with a half-life of 12 years, which does not occur significantly in nature. For use in this fusion reaction tritium must be made by reaction of the lithium isotope of mass 6 with a neutron:

$$^6Li + {}^1n \rightarrow {}^4He + {}^3T$$

Lithium Neutron Helium Tritium

Natural lithium contains 7.5 percent ^6Li.

The energy released in these two reactions may be calculated from the decrease in mass between the reactants and the products: